VS-09 Infrastructure and Platform Enablement

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1. Purpose and value created

VS-09 provides the technical and digital foundation for every part of the NSI. Where the other value streams focus on data, methods, or governance, VS-09 makes sure that all of these activities can actually run: safely, efficiently, and at scale. It delivers the IT environments, data platforms, tools, and integration services that statistical production needs, from collection to dissemination. It keeps systems secure, resilient, and compliant, while staying flexible enough to adopt new technologies. VS-09 describes the business view of this enablement: the capability of providing platforms and services. The detailed application and technology architecture is the subject of the application and technology layers of the SAF. The value created is **reliable enablement**: infrastructure becomes an active enabler of the statistical mission rather than a background function.

VS-09 is a supporting value stream. It aligns with the GAMSO areas *Capability Management* and *Corporate Support* rather than with the production phases of GSBPM. This section describes its business layer for all four process types.

- **Platform engineer or operations staff:** the stages in attribute 3, which describe the enablement life cycle.
- **Enterprise or platform architect:** the Design processes in attribute 4.
- **Process owners of other value streams:** attributes 2 and 8, which describe what VS-09 provides and what it needs from them.

- **Manager who wants the big picture:** attributes 1, 2, and 9. Examples and scenarios are part of the SAF Toolkit.

> **How a supporting value stream differs from a primary one**
>
> VS-09 has no steady-state data transition. It does not move data along the chain; it provides the environments in which the chain runs. It runs continuously as an always-on service, not per production cycle. Its output is an enabling capability rather than a data artifact: platforms, environments, integration services, and assured availability. The steady states and their quality gates are business agreements; VS-09 provides the storage, catalog, and event interfaces that implement them.

PLACEHOLDER: ArchiMate view VS-09 (a) value stream and capabilities

2. Position and interaction with the chain

VS-09 is not a stage in the production line. It is the foundation underneath it: every primary and supporting value stream runs on the platforms that VS-09 provides. VS-09 receives requirements that follow from business capabilities, data volumes, and analytical complexity. It receives specific requests from VS-08 for new technology and from VS-06 for dissemination platforms. It provides platforms, integration services, and operational assurance. Its security and continuity are governed by VS-07.

Counterpart	VS-09 provides	VS-09 receives
VS-01 to VS-05 (primary value streams)	Environments for collection, processing, and analysis; steady-state repositories; catalog; event interfaces	Requirements, capacity demand, incidents
VS-06 Dissemination and Service Delivery	Portals, APIs, dashboards, secure research environments, data space connectors	Requirements for channels and scaling, especially at release moments
VS-07 Governance and Compliance	Evidence on security posture and continuity	Security and continuity policy
VS-08 Innovation and Methodological Development	Research environments and production runtime for new technology	Requests to adopt new technology
VS-10 Customer and Stakeholder Engagement	Platforms for user services and communication	Requirements from users on availability and self-service
External parties: suppliers and service providers	Requirements and service agreements	Infrastructure, cloud services, software, and support

PLACEHOLDER: ArchiMate view VS-09 (c) interaction with the chain: VS-09 as the foundation underneath all value streams, providing environments, steady-state repositories, catalog, event interfaces, and dissemination platforms, and receiving requirements, capacity demand, and incidents; security and continuity governed by VS-07

3. Stages

The stages of VS-09 form a continuous enablement life cycle: define, design, provision, integrate, assure, and evolve. Requirements lead to architecture, architecture to platforms, platforms to services that are monitored and kept safe, and experience to modernization.

Stage	Name	Value added
VS-09.1	Define Digital and Technical Requirements	It is clear which technical capabilities the business needs
VS-09.2	Design Platform and Infrastructure Architecture	There is a coherent blueprint for storage, processing, integration, and access
VS-09.3	Provision and Configure Platforms	Controlled environments are available to the value streams
VS-09.4	Enable Integration and Orchestration Services	Processes and value streams are connected through stable interfaces
VS-09.5	Ensure Performance, Security, and Continuity	The chain runs reliably, safely, and can recover from disruption
VS-09.6	Evolve and Modernize the Technical Ecosystem	The foundation stays current and ready for new data and methods

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. Some activities have no reference process in the SAF; they are carried out through NSI-specific IT management processes, which the column shows as *NSI-specific*. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-09.1 Define Digital and Technical Requirements

Value created: *A clear view of the technical capabilities that the business needs.*

The life cycle begins with the requirements. VS-09 translates business capabilities, data volumes, and analytical complexity into technical requirements. It collects the requirements of the value streams for storage, processing, integration, access, and availability. It includes the requirements that follow from the steady states: immutable versions, persistent identifiers, field-level lineage, and the handover events. It adds the requirements of VS-07 for security, privacy, and continuity. Each requirement is linked to the business need it serves, so that platform choices can be explained in business terms.

Activity	Key outputs	SAF process	Other reference
Translate business capabilities and data volumes into technical requirements	Technical requirements	PRD-2.6 Design production systems and workflows	4.1.7 Architecture management
Collect the requirements of the value streams for storage, processing, integration, access, and availability	Requirements inventory	PRD-2.6 Design production systems and workflows	SAF-P05 IT and technology is everyone's business
Include the requirements of the steady states and of VS-07 Governance and Compliance	Requirements for steady states, security, and continuity	PRD-2.6 Design production systems and workflows	SAF-P07 Implement loosely coupled processes and systems; SAF-P10 Security by design
Link each requirement to the business need it serves	Traceability of requirements	NSI-specific	SAF-P04 Maximize the benefits for the organization

VS-09.2 Design Platform and Infrastructure Architecture

Value created: *A coherent blueprint for storage, processing, integration, and secure access.*

The requirements feed the design of the platform and infrastructure architecture. The blueprint covers data storage, processing, integration, and secure access, for example a data fabric, a data marketplace, and compute platforms. It decides which deployment model fits each component: on-premises, private cloud, public cloud, or a hybrid. It defines how the steady states are implemented on the platforms, so that each state has its repository, catalog entry, and access control. Shared platforms are preferred over solutions per product, and reuse comes before own development. The architecture is reviewed by the architecture board before it is realized.

Activity	Key outputs	SAF process	Other reference
Design the blueprint for storage, processing, integration, and access	Platform architecture	PRD-2.6 Design production systems and workflows	4.1.7 Architecture management
Decide the deployment model per component: on-premises, private, public, or hybrid cloud	Deployment decisions	PRD-2.6 Design production systems and workflows	3.2.5 IT infrastructure
Define how the steady states are implemented on the platforms	Implementation pattern for steady states	PRD-2.6 Design production systems and workflows	SAF-P07 Implement loosely coupled processes and systems **[VC29]**
Review the architecture with the architecture board	Architecture decision record	NSI-specific	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes

> **i** On the Medallion architecture
>
> A platform team may use Bronze, Silver, and Gold to organize storage, staging, and table optimization inside the platform. That is an engineering pattern, not a governance model. It describes how data has been refined, not what the data means, what methodological state it is in, or what another statistical process may rely on. The steady states remain the governance model; Medallion, if used at all, sits underneath it as a storage convention. **[VC29]**

VS-09.3 Provision and Configure Platforms

Value created: *Controlled environments that the value streams can use.*

The architecture is realized by provisioning and configuring the platforms. Environments are deployed for development, testing, research, and production, with the necessary governance, monitoring, and security controls. The secure processing environment for researchers is provisioned with its specific controls. Self-service is offered where possible, so that teams can obtain standard environments without waiting. Each environment is documented and registered, with an owner and a service level.

Activity	Key outputs	SAF process	Other reference
Deploy environments for development, testing, research, and production	Deployed environments	PRD-3.4 Configure workflows	3.2.5 IT infrastructure
Configure governance, monitoring, and security controls per environment	Controlled environments	PRD-3.4 Configure workflows	SAF-P10 Security by design
Provision the secure processing environment for researchers	Secure processing environment	MDS-4 Provide a secure processing environment	SAF-P11 Privacy by design

Activity	Key outputs	SAF process	Other reference
Offer standard environments as self-service and register them with an owner and service level	Self-service catalog, environment register	NSI-specific	SAF-P05 IT and technology is everyone's business; 3.4.4 Self-service

VS-09.4 Enable Integration and Orchestration Services

Value created: *Processes and value streams connected through stable, versioned interfaces.*

The platforms are connected through integration and orchestration services. These are the APIs, workflow engines, and automation that link the business services and processes. They include the steady-state event interfaces that the primary value streams rely on, such as *Raw Data Registered* and *Processed Data Registered*. They include the connectors for data spaces and for transmission to international organizations. Interfaces are versioned and documented, so that a change on one side does not break the other, in line with SAF-P07.

Activity	Key outputs	SAF process	Other reference
Provide APIs, workflow engines, and automation	Integration and orchestration services	PRD-3.4 Configure workflows	3.2.1 Application development
Provide the steady-state event interfaces of the primary value streams	Event interfaces	PRD-3.4 Configure workflows	SAF-P07 Implement loosely coupled processes and systems
Provide connectors for data spaces and international transmission	Data space connectors	DSP-4 Exchange data under usage control	4.1.6 Collaboration
Version and document every interface	Interface catalog	NSI-specific	SAF-P03 Active life cycle management

VS-09.5 Ensure Performance, Security, and Continuity

Value created: *A chain that runs reliably and safely and can recover from disruption.*

Day to day, VS-09 keeps the platforms running and safe. It monitors performance and capacity, with special attention to known peaks such as release moments. It implements the technical data protection that the classification of each dataset requires, including the storage of immutable versions. It manages incidents and restores service within the agreed times. It guarantees availability and recovery from disaster through tested continuity plans. Evidence on security posture and continuity is reported to VS-07.

Activity	Key outputs	SAF process	Other reference
Monitor performance and capacity, including known peaks	Performance and capacity reports	PRD-3.5 Test production systems	SAF-P02 Active support
Implement technical data protection and the storage of immutable versions	Protected storage	DLM-5 Store and version data	SAF-P10 Security by design; 3.3.6 IT security services
Manage incidents and restore service	Incident records	NSI-specific	3.3.3 Security incident management
Guarantee availability and disaster recovery through tested continuity plans	Continuity plans, test results	NSI-specific	SAF-P06 Business continuity

VS-09.6 Evolve and Modernize the Technical Ecosystem

Value created: *A foundation that stays current and ready for new data sources, methods, and ways of working.*

The life cycle closes with the modernization of the ecosystem. New technologies are introduced, often after they have been tested in VS-08. Components that are obsolete or no longer supported are retired according to a roadmap. Dependence on single suppliers is reviewed, in line with SAF-P01. The roadmap is aligned with the strategy of the NSI and with the architecture. Changes are planned so that the production value streams can continue without disruption.

Activity	Key outputs	SAF process	Other reference
Introduce new technologies transferred from VS-08 Innovation and Methodological Development	Adopted technologies	No reference process (NSI-specific)	4.1.3 Change management
Retire obsolete and unsupported components according to the roadmap	Retirement records	No reference process (NSI-specific)	SAF-P03 Active life cycle management
Review dependence on single suppliers	Supplier dependence review	No reference process (NSI-specific)	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Maintain the modernization roadmap, aligned with strategy and architecture	Modernization roadmap	No reference process (NSI-specific)	4.1.7 Architecture management

4. The four process types

VS-09 is itself a capability that must be designed, built, operated, and improved. The four process types therefore apply to VS-09 as they do to the primary value streams, but they govern the platform capability rather than a data pipeline. Design produces the operating model and the service catalog, Implementation builds the service management and the supplier arrangements, Execution runs the stages of attribute 3, and Management evaluates whether the platforms meet the needs of the value streams. **[VC26]**

Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. Where no reference process exists, the sub-process is marked as NSI-specific.

Design: provides the platform operating model

Design sub-process	Refines	Main outputs
Design the Platform Operating Model and Service Catalog	PRD-2.6 Design production systems and workflows; NSI-specific	Operating model, service catalog, service levels
Design the Security and Continuity Architecture	PRD-2.6 Design production systems and workflows; DLM-4 Classify and protect data	Security architecture, continuity requirements

Design the Platform Operating Model and Service Catalog

Refines: PRD-2.6 Design production systems and workflows; NSI-specific IT management processes.

This sub-process designs how VS-09 delivers its services. It defines the service catalog: which platforms and services are offered, to whom, and with which service levels. It defines the operating model: which services are run by the NSI, which are shared with other government bodies, and which are sourced from suppliers. It defines how teams request services and how requests are prioritized. It sets the service levels for availability, response, and recovery. The catalog makes the offer of VS-09 visible and comparable for all value streams.

Activity	Key outputs	SAF process	Other reference
Define the service catalog with services, users, and service levels	Service catalog	NSI-specific	3.4.1 Services management
Define the operating model: own, shared, and sourced services	Operating model	NSI-specific	3.1.2 Purchasing and contract management
Define the procedure for requests and their prioritization	Request procedure	PRD-2.6 Design production systems and workflows	SAF-P04 Maximize the benefits for the organization

Design the Security and Continuity Architecture

Refines: PRD-2.6 Design production systems and workflows; DLM-4 Classify and protect data.

This sub-process designs how platforms are protected and kept available. It translates the security and privacy policy of VS-07 into a security architecture: identity and access control, encryption, logging, and separation of environments. It links the protection level of each platform to the classification of the data it holds. It defines the continuity requirements per service, based on the impact of an outage on the chain and on release commitments. It designs the recovery approach and the tests that prove it works.

Activity	Key outputs	SAF process	Other reference
Translate the security and privacy policy into a security architecture	Security architecture	PRD-2.6 Design production systems and workflows	SAF-P10 Security by design; 3.3.2 Security planning
Link platform protection to the classification of the data held	Protection levels per platform	DLM-4 Classify and protect data	3.3.4 Logical identity and access control
Define continuity requirements and the recovery approach per service	Continuity requirements	NSI-specific	SAF-P06 Business continuity

Implementation: provides the service management

Implementation sub-process	Refines	Main outputs
Establish Service Management and Operations	NSI-specific	Service desk, incident and change management, monitoring
Arrange Supplier and Sourcing Agreements	NSI-specific	Contracts, service agreements with suppliers

Establish Service Management and Operations

Refines: NSI-specific IT service management processes.

This sub-process puts the operating model into practice. The service desk is set up for requests and incidents. Incident, problem, and change management are established, with clear roles and escalation paths. Monitoring of availability, performance, capacity, and security is configured across all platforms. Operational procedures and runbooks are written and tested. Operations staff are trained, so that services are supported actively and not only built once, in line with SAF-P02.

Activity	Key outputs	SAF process	Other reference
Set up the service desk for requests and incidents	Service desk	NSI-specific	3.4.1 Services management
Establish incident, problem, and change management	Service management processes	NSI-specific	SAF-P02 Active support
Configure monitoring and write operational procedures	Monitoring, runbooks	NSI-specific	3.2.3 Technical management

Arrange Supplier and Sourcing Agreements

Refines: NSI-specific procurement processes.

This sub-process arranges the services that VS-09 sources from outside. It selects suppliers according to the operating model and the procurement rules. It negotiates service agreements that match the service levels the NSI promises to its value streams. It includes requirements for security, data location, exit, and the use of open standards. It makes sure that the NSI keeps control over critical components and data, in line with SAF-P01. Agreements are registered and reviewed before they expire.

Activity	Key outputs	SAF process	Other reference
Select suppliers according to the operating model and procurement rules	Supplier selection	NSI-specific	3.1.2 Purchasing and contract management
Negotiate service agreements that match the promised service levels	Service agreements	NSI-specific	3.4.3 Service agreement
Include requirements for security, data location, exit, and open standards	Contract requirements	NSI-specific	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes

Execution: runs the platform services

Execution runs the stages of attribute 3. VS-09.1 and VS-09.2 run when needs change or new value streams are set up; VS-09.3 and VS-09.4 run per request and per release of the platforms; VS-09.5 runs continuously; VS-09.6 runs as a roadmap.

Aspect	VS-09
Trigger	*Platform or service request*; *Incident*; *Capacity change*; *Technology onboarding* from VS-08 Innovation and Methodological Development
Cadence	Continuous operation; per request; roadmap cycle
Performed by	ROL-11 Platform engineer, ROL-10 Data engineer, ROL-20 Security officer
Input	Requirements of the value streams, security and continuity policy, supplier services
Output	Platforms, environments, integration services, event interfaces, availability and continuity evidence
Control point	Architecture review before realization; acceptance of an environment before production use
Exception path	A major incident escalated according to the continuity plan; a security incident handled with ROL-20 Security officer and VS-07 Governance and Compliance

Management: provides service performance reports

Management sub-process	Refines	Main outputs
Monitor Service Performance	PRD-8.1 Gather evaluation inputs	Service level reports, capacity and security reports
Evaluate and Improve the Platform Capability	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Satisfaction of value streams, improvement plan

Monitor Service Performance

Refines: PRD-8.1 Gather evaluation inputs.

This sub-process tracks how the platforms perform against the agreed service levels. It measures the key performance indicators of VS-09 (see attribute 9), such as availability, time to restore service, and the reliability of event interfaces. It reports capacity headroom ahead of known peaks. It reports the security posture, together with VS-07. It reviews the performance of suppliers against their agreements. The results are shared with the process owners of the value streams that depend on the platforms.

Activity	Key outputs	SAF process	Other reference
Measure the key performance indicators of VS-09 Infrastructure and Platform Enablement	Service level reports	PRD-8.1 Gather evaluation inputs	SAF-P02 Active support
Report capacity headroom and security posture	Capacity and security reports	PRD-8.1 Gather evaluation inputs	SAF-P06 Business continuity; SAF-P10 Security by design
Review supplier performance against agreements	Supplier performance review	NSI-specific	3.4.3 Service agreement

Evaluate and Improve the Platform Capability

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This sub-process asks whether the platforms meet the needs of the value streams. It collects the satisfaction of the process owners with the services. It analyzes recurring incidents and their root causes. It assesses

the cost and value of the services. It feeds the findings into the roadmap of VS-09.6 and into the design of the service catalog. Improvements are prioritized by impact and feasibility and tracked, in line with SAF capability 4.1.4 *Improvement management*.

Activity	Key outputs	SAF process	Other reference
Collect the satisfaction of process owners with the services	Satisfaction survey	PRD-8.2 Conduct evaluation	3.4.1 Services management
Analyze recurring incidents and their root causes	Problem analysis	PRD-8.2 Conduct evaluation	
Feed findings into the roadmap and the service catalog, and track improvements	Improvement plan	PRD-8.3 Agree an action plan	4.1.4 Improvement management

5. Actors and roles

VS-09 brings architecture, engineering, operations, and security together. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the platform or cloud architect, the DevOps engineer, and service desk and operations staff are ROL-11 Platform engineer, the integration engineer is ROL-10 Data engineer, and the security and continuity officer is ROL-20 Security officer. The chief technology or infrastructure officer is part of the management of the NSI (ACT-01) and a member of the collaboration COL-07 *Architecture board*.

Actor or role	ID	Responsibility in VS-09 Infrastructure and Platform Enablement
National Statistical Institute	ACT-01	Its management, including the chief technology officer, is accountable for the technical foundation and its strategy
Supplier and service provider	ACT-09	Supplies infrastructure, cloud services, software, and support under service agreements
Process owners of all value streams	ROL-01	Define requirements and run their processes on the platforms
Data custodian	ROL-04	Operates storage, access control, retention, and key management on the platforms
Metadata and data architect	ROL-08	Aligns the catalog and repositories with the metadata model and the steady states
Data engineer	ROL-10	Builds APIs, workflow engines, orchestration, and event interfaces
Platform engineer	ROL-11	Designs the platform architecture; provisions, configures, and operates environments; runs incident management
Security officer	ROL-20	Implements security controls, monitoring, and disaster recovery, together with VS-07 Governance and Compliance

6. Capabilities, processes, and functions

Capability foundations

Capability group	Capabilities	Role in VS-09 Infrastructure and Platform Enablement
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management, 3.2.3 Technical management, 3.2.4 Digital workplace management, 3.2.5 IT infrastructure	Build, configure, and operate applications, platforms, and infrastructure
3.3 Security and privacy	3.3.2 Security planning, 3.3.3 Security incident management, 3.3.4 Logical identity and access control, 3.3.5 Physical security and access control, 3.3.6 IT security services	Protect platforms and data, and handle incidents
3.1 Business operations	3.1.2 Purchasing and contract management	Source infrastructure and services from suppliers
3.4 Customer and chain interaction	3.4.1 Services management, 3.4.3 Service agreement, 3.4.4 Self-service	Offer platform services with service levels and self-service
4.1 Strategy and leadership	4.1.3 Change management, 4.1.7 Architecture management	Manage technological change and the architecture of the platforms

Stages, processes, and functions

The table follows the mapping in *Business Processes, Mapping to value stream stages.*

Stage	Realized by processes	Main functions
VS-09.1 Define Digital and Technical Requirements	PRD-2.6 Design production systems and workflows	FUN-2.5 Manage enterprise architecture; FUN-3.1 Design and build statistical products
VS-09.2 Design Platform and Infrastructure Architecture	PRD-2.6 Design production systems and workflows	FUN-2.5 Manage enterprise architecture; FUN-10.3 Manage information technology
VS-09.3 Provision and Configure Platforms	PRD-3.4 Configure workflows; MDS-4 Provide a secure processing environment	FUN-8.3 Operate secure research environments; FUN-10.3 Manage information technology
VS-09.4 Enable Integration and Orchestration Services	PRD-3.4 Configure workflows; DSP-4 Exchange data under usage control	FUN-9.2 Operate data space connectivity; FUN-10.3 Manage information technology
VS-09.5 Ensure Performance, Security, and Continuity	PRD-3.5 Test production systems; DLM-5 Store and version data	FUN-10.3 Manage information technology; FUN-10.8 Manage information security and privacy
VS-09.6 Evolve and Modernize the Technical Ecosystem	No reference process (NSI-specific)	FUN-2.2 Develop capability improvements; FUN-10.3 Manage information technology

PLACEHOLDER: ArchiMate view VS-09 (b) stages realized by processes and functions

Framework mappings

Where the primary value streams map to the production phases of GSBPM, VS-09 maps to the capability and support activities of GAMSO.

GAMSO area	Relevance to VS-09 Infrastructure and Platform Enablement	SAF capabilities
Capability Management	Manage the IT capability; introduce and modernize technology	4.1.3 Change management, 4.1.7 Architecture management
Corporate Support	Manage IT services and operations, security, and business continuity	3.2.1 to 3.2.5, 3.3.2 to 3.3.6, 3.4.1 Services management

7. Principles

Principle	Statement	How VS-09 Infrastructure and Platform Enablement applies it
SAF-P01	Reuse before open source, open source before buying, buying before building	Favors open, shared platform components and open standards, to keep collective control and avoid dependence on single suppliers [**VC3**]
SAF-P02	Active support	Runs platforms as actively supported, monitored services, not one-off builds
SAF-P05	IT and technology is everyone's business	Provides self-service, well-documented platforms, so technology serves all teams
SAF-P06	Business continuity	Guarantees availability, resilience, and recovery from disaster for the whole chain
SAF-P07	Implement loosely coupled processes and systems	Provides the versioned interfaces and events that keep the value streams decoupled
SAF-P10	Security by design	Builds security and access control into every platform and environment from the start

8. Interfaces, services, and events

Business services

Service	Offered to	Realized in	Interfaces
Platform and environment service	All value streams	VS-09.3 Provision and Configure Platforms	Service catalog, self-service portal, service desk
Steady-state storage and catalog service	VS-01 to VS-06 (primary value streams)	VS-09.3 Provision and Configure Platforms; VS-09.5 Ensure Performance, Security, and Continuity	Repositories per steady state, data catalog

Service	Offered to	Realized in	Interfaces
Integration and event service	All value streams	VS-09.4 Enable Integration and Orchestration Services	APIs, workflow engines, event interfaces, data space connectors
Dissemination platform service	VS-06 Dissemination and Service Delivery	VS-09.3 Provision and Configure Platforms; VS-09.4 Enable Integration and Orchestration Services	Portals, APIs, dashboards, secure research environment
Research environment service	VS-08 Innovation and Methodological Development	VS-09.3 Provision and Configure Platforms	Sandboxes with research tooling
Operations and continuity service	All value streams, VS-07 Governance and Compliance	VS-09.5 Ensure Performance, Security, and Continuity	Monitoring, incident management, continuity reports

Business events

Event	Triggers or marks
Platform or service request	Start of VS-09.3 Provision and Configure Platforms or VS-09.4 Enable Integration and Orchestration Services for a request
Technology onboarding	Adoption of a technology transferred from VS-08 Innovation and Methodological Development in VS-09.6 Evolve and Modernize the Technical Ecosystem
Capacity change	Adjustment of capacity in VS-09.5 Ensure Performance, Security, and Continuity, for example ahead of a release
Incident / Service restored	Start and end of incident handling in VS-09.5 Ensure Performance, Security, and Continuity
Security or continuity review	Periodic review with VS-07 Governance and Compliance
Component retired	Retirement according to the roadmap in VS-09.6 Evolve and Modernize the Technical Ecosystem

PLACEHOLDER: VS-09 event choreography between the value streams, VS-09, suppliers, and VS-07: service request, provisioning, incident and restoration, capacity change, security review, technology onboarding, and retirement

9. Key performance indicators

KPI	Description	Measurement
Platform availability	Measured availability against the agreed service level, per platform	Monthly
Incident response	Mean time to restore service after an incident, per severity class	Per incident, trended quarterly
Event interface reliability	Delivery rate of the steady-state handover events that the chain depends on	Per interface, monthly
Capacity headroom	Share of platforms operating within their planned capacity	Monthly, ahead of known peaks

KPI	Description	Measurement
Security posture	Open critical vulnerabilities and time to patch, assessed with VS-07 Governance and Compliance	Continuous, reported quarterly
Modernization progress	Components retired or replaced according to the roadmap	Annually

10. Toolkit

Examples from NSI practice and a maturity checklist for VS-09 are part of the SAF Toolkit.

Toolkit: SAF-TK-V09 VS-09 Infrastructure and Platform Enablement companion [**VC17**]